MVA-PRISM v1

Rare Disease, Real Kid: MVA Hackathon 2026 — Track 1 Variant Prediction

Participant/team: BARYBIGAY
Proband: PROBAND01
Assembly: GRCh38
Source sample: WGS_EX2312012

Research-use statement. This submission is a hackathon research hypothesis, not a clinical diagnosis. The primary pair requires orthogonal confirmation, parental/long-range phasing, and functional evaluation of the missense allele.

1. Executive result

Rank	Gene / variant	VCF evidence	Interpretation
1	BUB1B chr15:40209701 T>G NM_001211.6:c.2210T>G p.Leu737Ter	0/1; DP 46; GQ 99; PASS	Pathogenic nonsense / loss-of-function
1	BUB1B chr15:40220612 T>G NM_001211.6:c.3006T>G p.Asn1002Lys	0/1; DP 28; GQ 99; PASS	Ultra-rare missense VUS; candidate second allele

The two alleles are approximately 10.9 kb apart. No phase-defining PGT/PID information was present for these two records, so the submission treats them as a candidate compound-heterozygous pair rather than proven variants in trans.

2. Phenotype used for prioritization

The supplied phenotype document lists rhabdomyosarcoma, nephrocalcinosis, short stature, failure to thrive, skeletal muscle atrophy, premature birth at 32 weeks, small for gestational age/severe low birth weight consistent with IUGR, and recurrent parental miscarriages. The challenge document specifically instructs participants to interpret these findings as a coherent cluster rather than optimize for a single feature.

3. Molecular rationale

BUB1B encodes BUBR1, a core spindle-assembly-checkpoint protein required for accurate chromosome segregation. Biallelic BUB1B loss/dysfunction causes mosaic variegated aneuploidy syndrome type 1 (MVA1), an autosomal-recessive chromosome-instability disorder with prenatal/postnatal growth restriction and substantial childhood cancer risk, including rhabdomyosarcoma.

The first allele, p.Leu737Ter, is a nonsense change expected to truncate/disrupt BUB1B. Current ClinVar records classify NM_001211.6:c.2210T>G (p.Leu737Ter), GRCh38 chr15:40209701 T>G, as pathogenic for MVA1. The second allele, p.Asn1002Lys, is a missense VUS. Public reproducibility analyses for this challenge report a CADD PHRED score of 24.5, SIFT deleterious, PolyPhen probably damaging, and extreme rarity in gnomAD v4. These computational data support prioritization but do not establish pathogenicity.

4. Workflow

1. Lock the phenotype vector from the challenge clinical document and preserve all supplied HPO terms.

2. Inspect the GRCh38 WGS VCF and require non-reference, PASS genotypes with usable DP/GQ.

3. Prioritize established MVA/chromosome-instability genes, especially BUB1B, CEP57 and TRIP13, while retaining a genome-wide counter-screen to reduce confirmation bias.

4. Annotate candidate consequence on the MANE transcript, evaluate population rarity and ClinVar evidence, and consider autosomal-recessive/biallelic models.

5. Rank candidate pairs by phenotype concordance, gene-disease validity, molecular consequence, rarity, inheritance plausibility, and technical evidence.

6. Separate secondary ClinVar screen hits from the primary causal hypothesis using finding_type=secondary.

5. Secondary findings retained for qualitative review

Gene	GRCh38 variant	EPCR	Comment
PRSS1	chr7:142750561 C>T	0.40	Screen hit; phenotype relevance not established
RBM8A	chr1:145927447 C>T	0.35	Needs disorder/inheritance review
FLG	chr1:152307547 G>A	0.30	Primary phenotype relevance not established
HK1	chr10:69315762 A>G	0.30	Needs variant-level clinical review
LZTR1	chr22:20996720 C>G	0.30	Primary phenotype relevance not established
GNRHR	chr4:67753920 C>T	0.25	Single heterozygous allele in recessive context

6. Limitations and confirmation

The principal limitation is phase: the two BUB1B variants are both heterozygous but are not proven to be in trans. The p.Asn1002Lys missense allele has no direct functional evidence in this analysis. A clinically meaningful conclusion would require orthogonal confirmation from an independent aliquot, parental segregation or validated long-read/linked phasing, and functional assessment of the missense allele. If the two alleles are in cis or the missense allele proves functionally neutral, the causal search should reopen to regulatory, structural, splice, and alternative-gene mechanisms.

7. AI/data-handling disclosure

OpenAI ChatGPT Team workspace, GPT-5.6 Sol, was used for literature synthesis, method drafting, cross-checking and artifact preparation. OpenAI states that business workspace inputs and outputs are not used to train models by default. Submitted coordinates/genotypes were directly verified against the supplied challenge VCF. Challenge data should be handled according to the organizers' access, retention and non-redistribution requirements.

8. Key references

• NCBI ClinVar. NM_001211.6(BUB1B):c.2210T>G (p.Leu737Ter), Variation ID 533901. https://www.ncbi.nlm.nih.gov/clinvar/variation/533901/

• Hanks S, et al. Constitutional aneuploidy and cancer predisposition caused by biallelic mutations in BUB1B. Nature Genetics. 2004;36:1159-1161. doi:10.1038/ng1449.

• Suijkerbuijk SJE, et al. Molecular causes for BUBR1 dysfunction in the human cancer predisposition syndrome mosaic variegated aneuploidy. Cancer Research. 2010;70:4891-4900. doi:10.1158/0008-5472.CAN-09-4319.

• ClinGen / NCBI gene-disease resources for BUB1B and MVA1. https://search.clinicalgenome.org/ and https://www.ncbi.nlm.nih.gov/gene/701

• OpenAI business data privacy. https://openai.com/business-data/